INCH-POUND

A–A–59590/7A
17 August 2005
SUPERSEDING
A-A-59590/7
24 September 2001

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .220 X .220 INCH BODY SIZE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59590	/07	B, C, H	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Options

Example: AA59590/07HJ48D is the PIN for a hard black anodize finished, 4.8 inch (121.9 mm) long card holder. The card holder also features two counterbored through holes for rivet mounting, and a combination of a lockwasher, flat washer, and screw self-lock element for added resistance to loosening.

SALIENT CHARACTERISTICS.

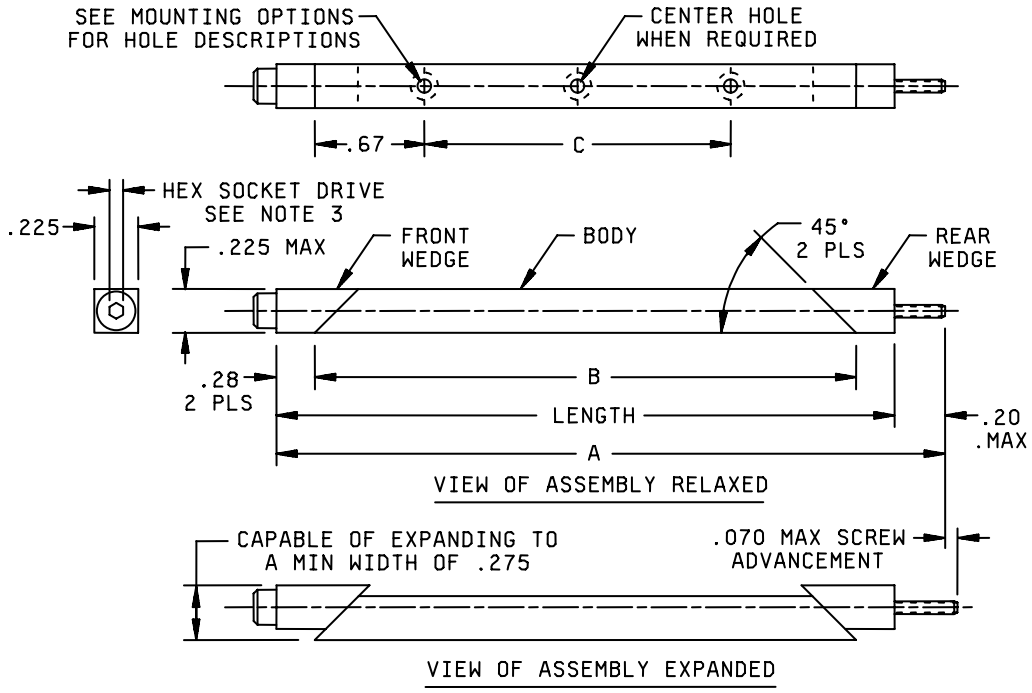
Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, in table I, and A–A–59590.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.070	1.78	.220	5.59	.225	5.72	.28	7.1
.20	5.1			.275	6.99	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.51 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.

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Finish. The finish options “B” (black anodize), “C” (chemical film), or “H” (hard black anodize) shall be as specified in A–A–59590.

Mounting. The mounting options shall be as specified in A–A–59590. The mounting options available for this specification sheet are as follows: “J” or “K” (rivet mounting holes), “N” (no mounting holes), “T” or “R” (tapped 0-80 UNF 2B holes), “V” or “S” (tapped 2-56 UNC 2B holes), or “G” or “M” (tapped M2.5 x.45 metric holes). Card holders using mounting option “J” or “K” may be shipped unassembled.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .160 to .175 inch (4.06 to 4.45 mm) deep.

Mounting holes (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension “LENGTH” ± .02 (0.5)	Dimension “A” (maximum)	Dimension “B” ± .03 (0.8)	Dimension “C” ± .005 (0.13)
18	1.8 (45.7)	2.10 (53.3)	1.24 (31.5)	.75 (19.1)
23	2.3 (58.4)	2.60 (66.0)	1.74 (44.2)	.40 (10.2)
28	2.8 (71.1)	3.10 (78.7)	2.24 (56.8)	.90 (22.9)
30	3.0 (76.2)	3.30 (83.8)	2.44 (62.0)	1.10 (27.9)
33	3.3 (83.8)	3.60 (91.4)	2.74 (69.6)	1.40 (35.6)
35	3.5 (88.9)	3.80 (96.5)	2.94 (74.7)	1.60 (40.6)
38	3.8 (96.5)	4.10 (104.1)	3.24 (82.3)	1.90 (48.3)
43	4.3 (109.2)	4.60 (116.8)	3.74 (95.0)	2.40 (61.0)
48	4.8 (121.9)	5.10 (129.5)	4.24 (107.7)	2.90 (73.7)
53	5.3 (134.6)	5.60 (142.2)	4.74 (120.4)	3.40 (86.4)
58	5.8 (147.3)	6.10 (154.9)	5.24 (133.1)	3.90 (99.1)
63	6.3 (160.0)	6.60 (167.6)	5.74 (145.8)	4.40 (111.8)
68	6.8 (172.7)	7.10 (180.3)	6.24 (158.5)	4.90 (124.5)
73	7.3 (185.4)	7.60 (193.0)	6.74 (171.2)	5.40 (137.2)
78	7.8 (198.1)	8.10 (205.7)	7.24 (183.9)	5.90 (149.9)
83	8.3 (210.8)	8.60 (218.4)	7.74 (196.6)	6.40 (162.6)
88	8.8 (223.5)	9.10 (231.1)	8.24 (209.3)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

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Options. Card holders can have the following options. Card holders not requiring the following option shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 2). The use of a lockwasher and flat washer located under the screw head will provide additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

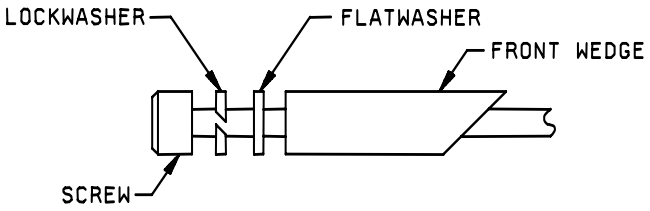


FIGURE 2. Lockwasher and flat washer details.

Screw self-locking element (see figure 3). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

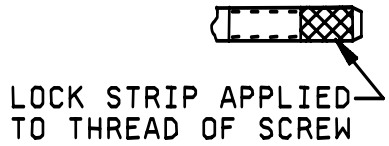


FIGURE 3. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	APW Electronic Solutions 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: sales.electronicsolutions@apw.com Uniform Resource Locator (URL): www.apw.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers in table II and on DSCC Drawing 84103 (table III) as shown. The CID PINs listed in table II are only for length designator "28". See table VI for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	42–6	CJ28	230–2.80H
CN28	42–6–A	CN28	230–2.80
CT28	42–6–T	CT28	230–2.80T0
CV28	42–6–S	CV28	230–2.80T2
CG28	42–6–M	CG28	M230–2.80TM2.5
CJ28E	42–6–L	CJ28E	230–2.80HL
CN28E	42–6–A–L	CN28E	230–2.80L
CT28E	42–6–T–L	CT28E	230–2.80T0L
CV28E	42–6–S–L	CV28E	230–2.80T2L
CG28E	42–6–M–L	CG28E	M230–2.80TM2.5L
CJ28W	42–6–LF	CJ28W	W230–2.80H
CN28W	42–6–LF–A	CN28W	W230–2.80
CT28W	42–6–LF–T	CT28W	W230–2.80T0
CV28W	42–6–LF–S	CV28W	W230–2.80T2
CG28W	42–6–LF–M	CG28W	MW230–2.80TM2.5

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 52094
CJ28D	42–6–LF–L	CJ28D	W230–2.80HL
CN28D	42–6–LF–A–L	CN28D	W230–2.80L
CT28D	42–6–LF–T–L	CT28D	W230–2.80T0L
CV28D	42–6–LF–S–L	CV28D	W230–2.80T2L
CG28D	42–6–LF–M–L	CG28D	MW230–2.80TM2.5L
BJ28	42–6–B	BJ28	A230–2.80H
BN28	42–6–B–A	BN28	A230–2.80
BT28	42–6–B–T	BT28	A230–2.80T0
BV28	42–6–B–S	BV28	A230–2.80T2
BG28	42–6–B–M	BG28	MA230–2.80TM2.5
BJ28E	42–6–B–L	BJ28E	A230–2.80HL
BN28E	42–6–B–A–L	BN28E	A230–2.80L
BT28E	42–6–B–T–L	BT28E	A230–2.80T0L
BV28E	42–6–B–S–L	BV28E	A230–2.80T2L
BG28E	42–6–B–M–L	BG28E	MA230–2.80TM2.5L
BJ28W	42–6–B–LF	BJ28W	WA230–2.80H
BN28W	42–6–B–LF–A	BN28W	WA230–2.80
BT28W	42–6–B–LF–T	BT28W	WA230–2.80T0
BV28W	42–6–B–LF–S	BV28W	WA230–2.80T2
BG28W	42–6–B–LF–M	BG28W	MWA230–2.80TM2.5
BJ28D	42–6–B–LF–L	BJ28D	WA230–2.80HL
BN28D	42–6–B–LF–A–L	BN28D	WA230–2.80L
BT28D	42–6–B–LF–T–L	BT28D	WA230–2.80T0L
BV28D	42–6–B–LF–S–L	BV28D	WA230–2.80T2L
BG28D	42–6–B–LF–M–L	BG28D	MWA230–2.80TM2.5L
HJ28	42–6–B3	HJ28	HA230–2.80H
HN28	42–6–B3–A	HN28	HA230–2.80
HT28	42–6–B3–T	HT28	HA230–2.80T0
HV28	42–6–B3–S	HV28	HA230–2.80T2
HG28	42–6–B3–M	HG28	MHA230–2.80TM2.5

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 52094
HJ28E	42–6–B3–L	HJ28E	HA230–2.80HL
HN28E	42–6–B3–A–L	HN28E	HA230–2.80L
HT28E	42–6–B3–T–L	HT28E	HA230–2.80T0L
HV28E	42–6–B3–S–L	HV28E	HA230–2.80T2L
HG28E	42–6–B3–M–L	HG28E	MHA230–2.80TM2.5L
HJ28W	42–6–B3–LF	HJ28W	WHA230–2.80H
HN28W	42–6–B3–LF–A	HN28W	WHA230–2.80
HT28W	42–6–B3–LF–T	HT28W	WHA230–2.80T0
HV28W	42–6–B3–LF–S	HV28W	WHA230–2.80T2
HG28W	42–6–B3–LF–M	HG28W	MWHA230–2.80TM2.5
HJ28D	42–6–B3–LF–L	HJ28D	WHA230–2.80HL
HN28D	42–6–B3–LF–A–L	HN28D	WHA230–2.80L
HT28D	42–6–B3–LF–T–L	HT28D	WHA230–2.80T0L
HV28D	42–6–B3–LF–S–L	HV28D	WHA230–2.80T2L
HG28D	42–6–B3–LF–M–L	HG28D	MWHA230–2.80TM2.5L

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Supersession information.

Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN
84103–38B	AA59590/07BJ28	84103–38C	AA59590/07CJ28
84103–39B	AA59590/07BJ30	84103–39C	AA59590/07CJ30
84103–40B	AA59590/07BJ35	84103–40C	AA59590/07CJ35
84103–41B	AA59590/07BJ38	84103–41C	AA59590/07CJ38
84103–42B	AA59590/07BJ48	84103–42C	AA59590/07CJ48
84103–43B	AA59590/07BJ53	84103–43C	AA59590/07CJ53
84103–44B	AA59590/07BJ58	84103–44C	AA59590/07CJ58

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TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/07	Vendor similar designator or type part number 1/ 2/ CAGE 52094
HJ18D	42–4–B3–LF–L	HJ18D	WHA230–1.80HL
HJ23D	42–5–B3–LF–L	HJ23D	WHA230–2.30HL
HJ28D	42–6–B3–LF–L	HJ28D	WHA230–2.80HL
HJ30D	3/	HJ30D	WHA230–3.00HL
HJ33D	42–7–B3–LF–L	HJ33D	WHA230–3.30HL
HJ35D	3/	HJ35D	WHA230–3.50HL
HJ38D	42–8–B3–LF–L	HJ38D	WHA230–3.80HL
HJ43D	42–9–B3–LF–L	HJ43D	WHA230–4.30HL
HJ48D	42–10–B3–LF–L	HJ48D	WHA230–4.80HL
HK53D	42–11–B3–LF–L	HK53D	WHA230–5.30EHL
HK58D	42–12–B3–LF–L	HK58D	WHA230–5.80EHL
HK63D	42–13–B3–LF–L	HK63D	WHA230C6.30EH
HK68D	42–14–B3–LF–L	HK68D	WHA230–6.80EHL
HK73D	42–15–B3–LF–L	HK73D	WHA230–7.30EHL
HK78D	42–16–B3–LF–L	HK78D	WHA230–7.80EHL
HK83D	42–17–B3–LF–L	HK83D	WHA230–8.30EHL
HK88D	42–18–B3–LF–L	HK88D	WHA230–8.80EHL

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Army – MI
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2005–006

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